

Striver's A to Z Sheet

DSA Learning Roadmap & Problem List

1. LOGIC BUILDING (Basics + Pattern Problems)

Fundamental problems to build a strong foundation in programming logic.

- ▷ Print all patterns (square, triangle, pyramid, diamond, etc.)
- ▷ Count digits in a number
- ▷ Reverse a number
- ▷ Check palindrome number
- ▷ GCD / HCF of two numbers
- ▷ Armstrong number
- ▷ Check prime number
- ▷ Print divisors
- ▷ Fibonacci series
- ▷ Recursion basics (print N times, reverse printing)

2. ARRAYS (Easy → Hard)



- ▷ Largest element in array
- ▷ Second largest element
- ▷ Check if array is sorted
- ▷ Remove duplicates from sorted array

- ▷ Left rotate array by 1 place
- ▷ Left rotate array by D places
- ▷ Move zeroes to end
- ▷ Linear search

Medium

- ▷ Union of two sorted arrays
- ▷ Missing number
- ▷ Maximum consecutive ones
- ▷ Single number
- ▷ Longest subarray with sum K
- ▷ Two Sum problem
- ▷ Sort 0s, 1s, 2s (Dutch flag)
- ▷ Majority element
- ▷ Kadane's Algorithm (max subarray)
- ▷ Rearrange array (positive/negative)
- ▷ Next permutation
- ▷ Leaders in array

Hard

- ▷ Longest consecutive sequence
- ▷ Set matrix zeroes
- ▷ Rotate matrix
- ▷ Spiral matrix
- ▷ Count subarrays with given sum

- ▷ Majority element II
- ▷ 3Sum problem
- ▷ 4Sum problem
- ▷ Largest subarray with 0 sum
- ▷ Merge overlapping intervals
- ▷ Count inversions

3. RECURSION (Pattern-wise)

● Basic Recursion

- ▷ Print name N times
- ▷ Print numbers from 1 to N
- ▷ Print numbers from N to 1
- ▷ Sum of first N numbers
- ▷ Factorial

● Intermediate

- ▷ Reverse an array using recursion
- ▷ Check palindrome (recursion)
- ▷ Fibonacci using recursion

● Backtracking / Advanced

- ▷ Subsequence generation
- ▷ Subset sum
- ▷ Combination Sum

▷ Combination Sum II

▷ Permutations

▷ N-Queens problem

▷ Sudoku solver

▷ Rat in a maze

▷ Palindrome partitioning

▷ K-th permutation

4. STRINGS (Basic → Medium)

● Basic

▷ Remove outermost parentheses

▷ Reverse words in a string

▷ Largest odd number in string

▷ Longest common prefix

▷ Isomorphic strings

▷ Check rotation

● Medium

▷ Frequency of characters

▷ Sort characters by frequency

▷ Longest substring without repeating characters

▷ Roman to integer

▷ String to integer (atoi)

▷ Longest palindromic substring

- ▷ Count substrings
-

Advanced

- ▷ Minimum insertions to make palindrome
-
- ▷ Z-function / pattern matching
-